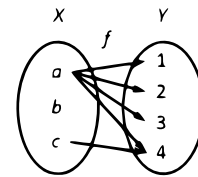


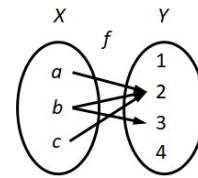
1. Let $X = \{a, b, c\}$ and $Y = \{1, 2, 3, 4\}$
- (i) For each of the following, give an example of a relation $f \subseteq X \times Y$ satisfying the statement:
- (a) $\exists x \in X \forall y \in Y (x, y) \in f$ (b)* $\exists y \in Y \forall x \in X (x, y) \in f$
(c)* $\forall y \in Y \exists x \in X (x, y) \in f$ (d) $\forall x \in X \forall x' \in X \forall y \in Y x \neq x' \rightarrow ((x, y) \notin f \vee (x', y) \notin f)$
- (ii) Are your examples in (i) functions?

Solution:

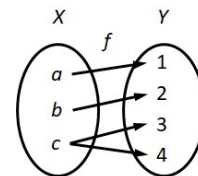
- (i) (a) $\exists x \in X \forall y \in Y (x, y) \in f$
 $f = \{(a, 1), (a, 2), (a, 3), (a, 4), (c, 4)\}$



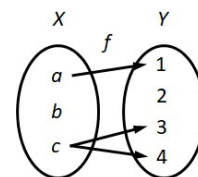
- (b)* $\exists y \in Y \forall x \in X (x, y) \in f$
 $f = \{(a, 2), (b, 2), (b, 3), (c, 2)\}$



- (c)* $\forall y \in Y \exists x \in X (x, y) \in f$
 $f = \{(a, 1), (b, 2), (c, 3), (c, 4)\}$



- (d) $\forall x \in X \forall x' \in X \forall y \in Y x \neq x' \rightarrow ((x, y) \notin f \vee (x', y) \notin f)$
 $f = \{(a, 1), (c, 3), (c, 4)\}$



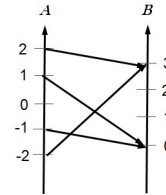
- (ii) All the above f are not functions.

Note to tutors: The students are likely to have different answers;
encourage them to come up with their own examples.

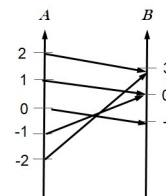
2. Let A and B be nonempty subsets of C , where $C = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$, and let $R = \{(x, y) \in A \times B \mid y = x^2 - 1\}$.
- Suppose $A = \{-2, -1, 0, 1, 2\}$ and $B = \{0, 1, 2, 3\}$. Is R a function?
 - Suppose $A = \{-2, -1, 0, 1, 2\}$.
Give an example of B such that $B \subseteq C$ and R is an onto function.
 - Suppose $B = \{0, 1, 2, 3\}$.
Give an example of A such that $A \subseteq C$ and R is a 1-1 function.

Solution: $C = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$, $A \subseteq C$, $B \subseteq C$, and $R = \{(x, y) \in A \times B \mid y = x^2 - 1\}$.

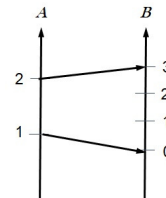
- (i) $A = \{-2, -1, 0, 1, 2\}$, $B = \{0, 1, 2, 3\}$. Not a function (no $(0, y) \in R$).



- (ii) $A = \{-2, -1, 0, 1, 2\}$ and $B = \{-1, 0, 3\}$ make R an onto function.



- (iii) $A = \{1, 2\}$ and $B = \{0, 1, 2, 3\}$ make R a one-one function.
(**Alternatives:** $A = \emptyset$, $A = \{-1\}$, etc.)



3. Recall that we define a *one-one* function $f : X \rightarrow Y$ as
“for all x_1 and x_2 in X , $x_1 = x_2$ if $f(x_1) = f(x_2)$ ”.

Explain the difference (if any) between this condition and

- * “for any x_1 and x_2 in X , $f(x_1) = f(x_2)$ whenever $x_1 = x_2$ ”;
- * “for every x_1 and x_2 in X , $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ ”;
- “there are no elements x_1 and x_2 in X such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$ ”;
- “ $x_1 \neq x_2$ and $f(x_1) = f(x_2)$ for some x_1 and x_2 in X ”;

Solution:

1-1 function: $\forall x_1 \in X \forall x_2 \in X f(x_1) = f(x_2) \rightarrow x_1 = x_2$

- (i)* “for any x_1 and x_2 in X , $f(x_1) = f(x_2)$ whenever $x_1 = x_2$ ”;
 $\forall x_1 \in X \forall x_2 \in X x_1 = x_2 \rightarrow f(x_1) = f(x_2)$ — converse (**not** the same)
 This statement doesn’t say anything useful: if $x_1 = x_2$, then of course $f(x_1) = f(x_2)$.
- (ii)* “for every x_1 and x_2 in X , $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ ”;
 $\forall x_1 \in X \forall x_2 \in X x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$ — contrapositive (the same)
- (iii) “there are no elements x_1 and x_2 in X such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$ ”
 $\sim \exists x_1 \in X \exists x_2 \in X x_1 \neq x_2 \wedge f(x_1) = f(x_2) \equiv \forall x_1 \in X \forall x_2 \in X x_1 = x_2 \vee f(x_1) \neq f(x_2)$
 $\equiv \forall x_1 \in X \forall x_2 \in X x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$ — same as (b)
- (iv) “ $x_1 \neq x_2$ and $f(x_1) = f(x_2)$ for some x_1 and x_2 in X ”
 $\exists x_1 \in X \exists x_2 \in X x_1 \neq x_2 \wedge f(x_1) = f(x_2)$ — **not** the same (in fact, **not** 1-1)

4. Recall that we define an *onto* function $f : X \rightarrow Y$ as

“given any y in Y , it is possible to find an x in X such that $y = f(x)$ ”.

Explain the difference (if any) between this condition and

- (i)* “for every x in X , there is y in Y such that $y = f(x)$ ”;
- (ii)* “there is x in X such that $y = f(x)$ for any y in Y ”;
- (iii)* “there is some y in Y such that $y = f(x)$ for some x in X ”;
- (iv) “there is some y in Y such that $y = f(x)$ for any x in X ”;
- (v) “there is no y in Y such that $y \neq f(x)$ for any x in X ”;

Solution:

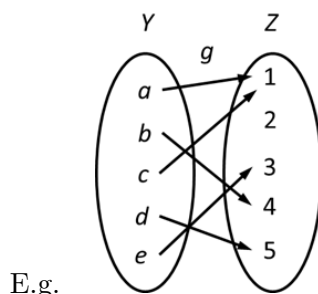
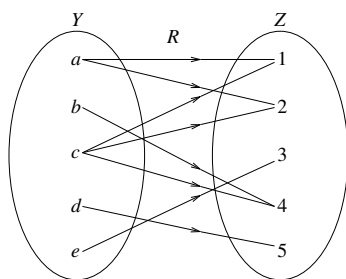
onto function: $\forall y \in Y \exists x \in X y = f(x)$

- (i)* “for every x in X , there is y in Y such that $y = f(x)$ ”;
 $\forall x \in X \exists y \in Y y = f(x)$ — **not** the same (actually, this is F1)
- (ii)* “there is x in X such that $y = f(x)$ for any y in Y ”;
 $\exists x \in X \forall y \in Y y = f(x)$ — **not** the same (e.g. $i_{\{a,b\}}$ is onto, but does not satisfy this)
- (iii)* “there is some y in Y such that $y = f(x)$ for some x in X ”;
 $\exists y \in Y \exists x \in X y = f(x)$ — **not** the same
- (iv) “there is some y in Y such that $y = f(x)$ for any x in X ”;
 $\exists y \in Y \forall x \in X y = f(x)$ — **not** the same
- (v) “there is no y in Y such that $y \neq f(x)$ for any x in X ”;
 $\sim \exists y \in Y \forall x \in X y \neq f(x) \equiv \forall y \in Y \exists x \in X y = f(x)$ — the same

Note to tutors: For Problems 3 and 4, you need to be prepared with some examples.

- 5.* Recall the relation R in Tutorial 4, Problem 5. Give an example of a function $g \subseteq R$ such that g is not one-one and not onto.

Solution:

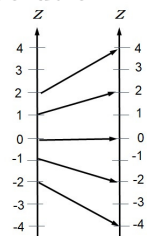


E.g.

g is a function that is not 1-1 and not onto.

6. Consider functions f and g from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(n) = 2n$ and $g(n) = \lfloor \frac{n}{2} \rfloor$. Which of the functions $f, g, g \circ f, f \circ g, f \circ f$ are 1-1? Which are onto? Determine the range for each function.

Solution:



f

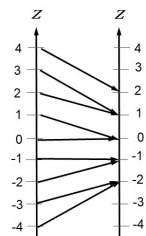
1-1

$$f(n) = f(k)$$

$$\Rightarrow 2n = 2k$$

$$\Rightarrow n = k$$

not onto



g

not 1-1

onto

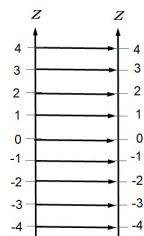
for any $y \in \mathbb{Z}$

pick $x = 2y$; then

$$g(x) = \lfloor \frac{x}{2} \rfloor$$

$$= \lfloor y \rfloor$$

$$= y$$



$g \circ f$

1-1

$$g \circ f(n) = g \circ f(k)$$

$$\Rightarrow g(2n) = g(2k)$$

$$\Rightarrow \lfloor \frac{2n}{2} \rfloor = \lfloor \frac{2k}{2} \rfloor$$

$$\Rightarrow n = k$$

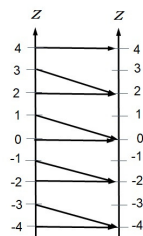
onto

for any $y \in \mathbb{Z}$,

pick $x = y$; then

$$g \circ f(x) = g(f(x))$$

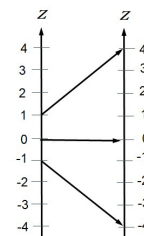
$$= g(2x) = \lfloor \frac{2x}{2} \rfloor = x$$



$f \circ g$

not 1-1

not onto



$f \circ f$

1-1

$$f \circ f(n) = f \circ f(k)$$

$$\Rightarrow f(2n) = f(2k)$$

$$\Rightarrow 4n = 4k$$

$$\Rightarrow n = k$$

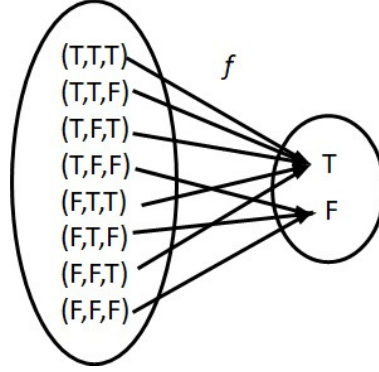
not onto

$\text{range}(f) = \{\text{even integers}\}$, $\text{range}(g) = \mathbb{Z}$, $\text{range}(g \circ f) = \mathbb{Z}$, $\text{range}(f \circ g) = \{\text{even integers}\}$, $\text{range}(f \circ f) = \{\text{integer multiples of 4}\}$.

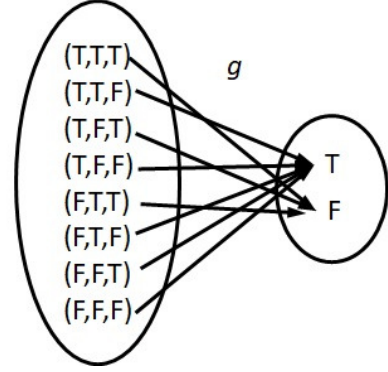
7. (i) Define Boolean functions f and g from $\{T, F\}^3$ to $\{T, F\}$ so that for all p, q and r ,
 (a)* $f(p, q, r) = (p \wedge q) \vee r$ (b) $g(p, q, r) = (p \vee q) \rightarrow \sim r$
 (ii) Using only $\sim$ and $\wedge$, define Boolean functions f' and g' such that $f = f'$ and $g = g'$.
 (iii) Suppose Q is a Boolean expression with n statement variables, and $f : \{T, F\}^n \rightarrow \{T, F\}$ is a Boolean function equivalent to Q . If f is not onto, what can you say about Q ?

Solution:

- (i) Define



(a)* $f(p, q, r) = (p \wedge q) \vee r$



(b) $g(p, q, r) = (p \vee q) \rightarrow \sim r$

- (ii) $(p \wedge q) \vee r \equiv \sim \sim ((p \wedge q) \vee r) \equiv \sim (\sim (p \wedge q) \wedge \sim r) \stackrel{\text{defn}}{=} f'(p, q, r)$
 $(p \vee q) \rightarrow \sim r \equiv \sim \sim ((\sim p \wedge \sim q) \vee \sim r) \equiv \sim (\sim (\sim p \wedge \sim q) \wedge r) \stackrel{\text{defn}}{=} g'(p, q, r)$
- (iii) Since f is not onto $\{T, F\}$, either $\forall \alpha \in \{T, F\}^n f(\alpha) = T$ or $\forall \alpha \in \{T, F\}^n f(\alpha) = F$
 i.e. Q is a tautology, or Q is a contradiction.

8. Let X and Y be sets and let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ be functions such that $g \circ f = i_X$. Prove that if f is onto, then g is one-one.

Solution:

Assume f is onto. Suppose $g(y) = g(y')$.

Since f is onto, there exist $x, x' \in X$ such that $y = f(x)$ and $y' = f(x')$.

Now, $x = i_X(x) = g \circ f(x) = g(f(x)) = g(y) = g(y') = g(f(x')) = g \circ f(x') = i_X(x') = x'$
 so $y = f(x) = f(x') = y'$.

Since $g(y) = g(y')$ implies $y = y'$, g is 1-1.

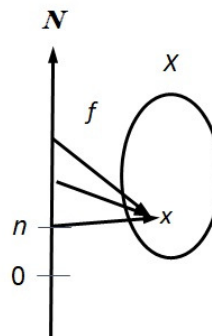
9.* Let X be a nonempty set and consider any function f from $\mathbb{N}$ to X that is onto. Define a function $g : X \rightarrow \mathbb{N}$ thus: $g(x)$ is the smallest integer n such that $f(n) = x$. Show that g is well-defined; i.e. show that

$\{(x, n) \in X \times \mathbb{N} \mid n \text{ is the smallest integer such that } f(n) = x\}$ is a function.

What is $f \circ g$? Is $g \circ f = i_{\mathbb{N}}$?

Solution:

An onto function $f : \mathbb{N} \rightarrow X$, $g = \{(x, n) \in X \times \mathbb{N} \mid n \text{ is the smallest integer such that } f(n) = x\}$



For each $x \in X$, let $N_x = \{n \in \mathbb{N} \mid f(n) = x\}$; note that $N_x \subseteq \mathbb{Z}$

Since $X \neq \emptyset$ and f is onto, $N_x \neq \emptyset$.

Also, N_x is bounded below by 0, i.e. $\forall n \in N_x, n \geq 0$.

By the Well-Ordering Principle, N_x has a smallest element $n_x \in N_x$, i.e. $\forall n \in N_x, n \geq n_x$.

Thus, $(x, n_x) \in g$ (F1: $\forall x \in X \exists n_x \in \mathbb{N} (x, n_x) \in g$).

Suppose $(x, n_x) \in g$ and $(x, n'_x) \in g$.

Then $f(n_x) = x$ and $f(n'_x) = x$, so $n_x \in N_x$ and $n'_x \in N_x$,

and $n_x \geq n'_x$ and $n'_x \geq n_x$ (by definition of g).

Thus, $n_x = n'_x$ (F2: $\forall x \in X \forall n_x \in \mathbb{N} \forall n'_x \in \mathbb{N} (x, n_x) \in g \wedge (x, n'_x) \in g \rightarrow n_x = n'_x$).

Hence, g is a function.

Moreover, for any $x \in X$, $f \circ g(x) = f(g(x)) = f(n_x) = x = i_X(x)$, so $f \circ g = i_X$.

The above shows that we can construct an “inverse” g for an onto function f , in the sense that $f \circ g = i_X$.

However, the following shows that this “inverse” may not satisfy $g \circ f = i_{\mathbb{N}}$.

Consider any f that is not 1-1 (e.g. $f : \mathbb{N} \rightarrow \mathbb{N}$ where $f(n) = \lfloor \frac{n}{2} \rfloor$ is an onto function).

Then for some $n \neq n'$, $f(n) = f(n') = x$ say. If $g \circ f = i_{\mathbb{N}}$, then

$n = i_{\mathbb{N}}(n) = g \circ f(n) = g(f(n)) = g(x) = g(f(n')) = g \circ f(n') = i_{\mathbb{N}}(n') = n'$, contradicting $n \neq n'$.

Therefore, $g \circ f \neq i_{\mathbb{N}}$ if f is not 1-1.

10.* The concept of an inverse function is central to cryptography.

- (i) Julius Caesar used a cipher that encrypts the message “attack today” as “dwwdfn wrgdb”. Define an encryption function E and its decryption function D (i.e. E^{-1}) for this cipher.
- (ii) The Caesar cipher is easy to break. It can be strengthened with a *key*. For example, if the key is “coolcat”, we drop the repeated letters to get “colat”, then use it and the rest of the alphabet to construct a rectangle:

<i>c</i>	<i>o</i>	<i>l</i>	<i>a</i>	<i>t</i>
<i>b</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>m</i>
<i>n</i>	<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>y</i>
<i>z</i>				

This rectangle can be used to define encryption and decryption functions so that “attack today” is encrypted as “crrchv rqnscs”. Define D and E for this cipher.

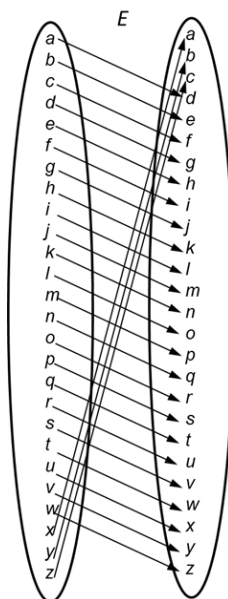
(The difference now is: Even if you know the encryption method, you still need to know the key to do the decryption; thus, D and E here are functions D_{coolcat} and E_{coolcat} that depend on the key “coolcat”.)

- (iii) Actually, we don’t need $D = E^{-1}$; we just need $D(E(x)) = x$ for all x . Construct two functions $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $g \circ f = i_X$ but $f \circ g \neq i_Y$.

Solution:

- (i) E is as shown in the arrow diagram.

E is bijective, so E^{-1} is a bijection, and $E \circ E^{-1} = i = E^{-1} \circ E$.



- (ii)

	α		β							
<i>a</i>	<i>g</i>	<i>l</i>	<i>q</i>	<i>v</i>		<i>c</i>	<i>o</i>	<i>l</i>	<i>a</i>	<i>t</i>
<i>b</i>	<i>h</i>	<i>m</i>	<i>r</i>	<i>w</i>		<i>b</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>c</i>	<i>i</i>	<i>n</i>	<i>s</i>	<i>x</i>		<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>m</i>
<i>d</i>	<i>j</i>	<i>o</i>	<i>t</i>	<i>y</i>		<i>n</i>	<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
<i>e</i>	<i>k</i>	<i>p</i>	<i>u</i>	<i>z</i>		<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>y</i>
<i>f</i>						<i>z</i>				

$\xrightarrow{E}$

$E(\alpha) = \beta$ and $D(\beta) = \alpha$, where α and β are in the same position.

- (iii) In Problem 6, $g \circ f = i_X$, but $f \circ g \neq i_Y$, so $g \neq f^{-1}$. (In fact, f^{-1} is not a function.)

11. We will prove that all students have the same sex.

Claim: There is only one sex among any group of n students, for any positive integer n .

Proof: By induction on n .

Basis $n = 1$: Since there is only 1 student, there is only 1 sex.

Induction Hypothesis: Suppose the claim is true if $n = k$, for some $k \geq 1$.

Induction Step: Consider any set S of $k + 1$ students. Remove one student x , so $S - \{x\}$ has k students. By the induction hypothesis, all students in $S - \{x\}$ have the same sex. Now put back the student and remove another student y , so $S - \{y\}$ has k students. Again, all students in $S - \{y\}$ have the same sex.

Thus, x and y have the same sex as students in $S - \{x, y\}$, so all students in S have the same sex. $\square$

What's wrong with this "proof"?

[The point here is: You can "prove" nonsense with bad logic, even if you stick to the structure provided by a proof technique.]

Solution:

Error: $S - \{x, y\}$ may be empty, i.e. the argument fails when $|S| = 2$.

Indeed, the induction step is bogus when S consists of one male and one female.